

MH2500 PROBABILITY AND INTRO TO STATISTICS

NANYANG TECHNOLOGICAL UNIVERSITY

Review Questions *without* solutions

Part 4

These are review questions for Chapter 7 and Chapter 8 in Ross' book.

- (1) Suppose that X and Y are both Bernoulli random variables. Show that X and Y are independent if and only if $\text{Cov}(X, Y) = 0$.
- (2) The nine players on a basketball team consist of 2 centers, 3 forwards, and 4 backcourt players. If the players are paired up at random into three groups of size 3 each, find (a) the expected value and (b) the variance of the number of triplets consisting of one of each type of player.
- (3) If a die is to be rolled until all sides have appeared at least once, find the expected number of times that outcome 1 appears.
- (4) A deck of n cards numbered 1 through n , initially in any arbitrary order, is shuffled in the following manner: At each stage, we randomly choose one of the cards and move it to the front of the deck, leaving the relative positions of the other cards unchanged. This procedure is continued until all but one of the cards has been chosen. At this point it follows by symmetry that all $n!$ possible orderings are equally likely. Find the expected number of stages that are required.
- (5) A deck of 52 cards is shuffled and a bridge hand of 13 cards is dealt out. Let X and Y denote, respectively, the number of aces and the number of spades in the hand.
 - (a) Show that X and Y are uncorrelated.
 - (b) Are they independent?
- (6) Let X be the length of the initial run in a random ordering of n ones and m zeroes. That is, if the first k values are the same (either all ones or all zeroes), then $X \geq k$. Find $\mathbb{E}(X)$.
- (7) A deck of $2n$ cards consists of n red and n black cards. The cards are shuffled and then turned over one at a time. Suppose that each time a red card is turned over, we win 1 unit if more red cards than black cards have been turned over by that time. (For instance, if $n = 2$ and the result is $rbrb$, then we would win a total of 2 units.) Find the expected amount that we win.

- (8) Suppose that it is known that the number of items produced in a factory during a week is a random variable with mean 50.
- (a) What can be said about the probability that this week's production will exceed 75?
 - (b) If the variance of a week's production is known to equal 25, then what can be said about the probability that this week's production will be between 40 and 60?
- (9) Suppose that the number of units produced daily at factory A is a random variable with mean 20 and standard deviation 3 and the number produced at factory B is a random variable with mean 18 and standard deviation 6. Assuming independence, derive an upper bound for the probability that more units are produced today at factory B than at factory A .
- (10) An astronomer is interested in measuring the distance, in light-years, from his observatory to a distant star. Although the astronomer has a measuring technique, he knows that, because of changing atmospheric conditions and normal error, each time a measurement is made it will not yield the exact distance, but merely an estimate. As a result, the astronomer plans to make a series of measurements and then use the average value of these measurements as his estimated value of the actual distance. If the astronomer believes that the values of the measurements are independent and identically distributed random variables having a common mean d (the actual distance) and a common variance of 4 (light-years), how many measurements need he make to be reasonably sure that his estimated distance is accurate to within $\pm .5$ lightyear?